

# **RAJALAKSHMI ENGINEERING COLLEGE**

**An Autonomous Institution, Affiliated to Anna University**

**Rajalakshmi Nagar, Thandalam – 602 105**

**Web Technology And Mobile Application**

**Experiment 11**

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## AIM:

To develop an Android application that writes user data into a file stored in external storage (SD card).

## ALGORITHM:

- Create a new Android project in Android Studio.
- Add permission in AndroidManifest.xml:
  - · <uses-permission  
android:name="android.permission.WRITE\_EXTERNAL\_STORAGE"/>
  - · · For Android 6.0+, request runtime permission in code.
- Design UI with:
  - EditText (for input)
  - Button (to save data)
- Use Environment.getExternalStorageDirectory() or getExternalFilesDir() to access storage.
- Create a file and write data using FileOutputStream.
- Show confirmation using Toast message.

## CODE:

### XML:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:padding="20dp"
    android:layout_width="match_parent"
    android:layout_height="match_parent">
```

```
<EditText
    android:id="@+id/editText"
    android:hint="Enter data"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"/>
```

```
<Button
    android:id="@+id/saveBtn"
    android:text="Save to SD Card"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"/>
</LinearLayout>
```

## MainActivity.java:

```
package com.example.sdcardwrite;
```

```
import androidx.appcompat.app.AppCompatActivity;  
import androidx.core.app.ActivityCompat;  
import android.Manifest;  
import android.content.pm.PackageManager;  
import android.os.Bundle;  
import android.os.Environment;  
import android.view.View;  
import android.widget.Button;  
import android.widget.EditText;  
import android.widget.Toast;
```

```
import java.io.File;  
import java.io.FileOutputStream;
```

```
public class MainActivity extends AppCompatActivity {
```

```
    EditText editText;  
    Button saveBtn;
```

```
    @Override  
    protected void onCreate(Bundle savedInstanceState) {  
        super.onCreate(savedInstanceState);  
        setContentView(R.layout.activity_main);
```

```
        editText = findViewById(R.id.editText);  
        saveBtn = findViewById(R.id.saveBtn);
```

```
        // Request permission  
        ActivityCompat.requestPermissions(this,  
            new String[]{Manifest.permission.WRITE_EXTERNAL_STORAGE}, 1);
```

```
        saveBtn.setOnClickListener(new View.OnClickListener() {  
            @Override  
            public void onClick(View view) {  
                writeToSDCard(editText.getText().toString());  
            }  
        });  
    }
```

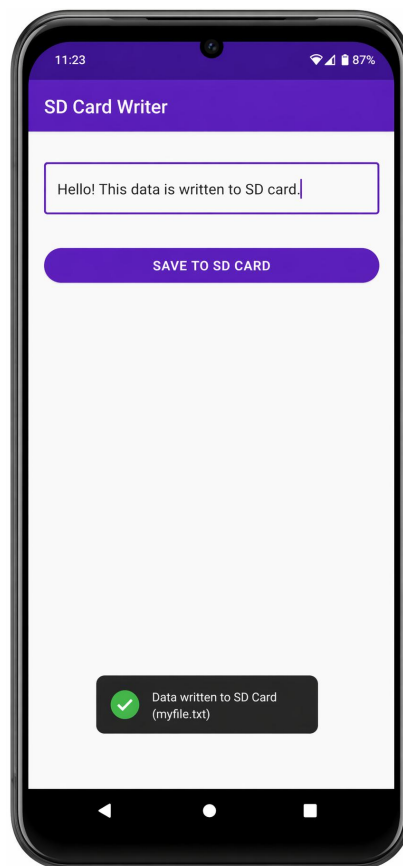
```
    private void writeToSDCard(String data) {  
        try {  
            File dir = Environment.getExternalStorageDirectory();  
            File file = new File(dir, "myfile.txt");
```

```
            FileOutputStream fos = new FileOutputStream(file);  
            fos.write(data.getBytes());  
            fos.close();
```

```
            Toast.makeText(this, "Data written to SD Card", Toast.LENGTH_SHORT).show();
```

```
} catch (Exception e) {  
    Toast.makeText(this, "Error: " + e.getMessage(), Toast.LENGTH_SHORT).show();  
}  
}
```

## OUTPUT:



## RESULT:

The application successfully writes user-entered data into a file (myfile.txt) stored in the device's external storage (SD card), and displays a confirmation message.